

PDF 2: Introduction to Deep Learning

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses artificial neural networks to learn from large amounts of data. Deep learning models are designed to automatically discover patterns and features from raw data without requiring manual feature extraction.

Deep Learning is inspired by the structure and function of the human brain. It uses multiple layers of artificial neurons to process information.

Neural Networks

A neural network is a computational model made up of connected nodes called neurons. These neurons work together to learn patterns from data.

A basic neural network contains:

1. Input Layer

The input layer receives data from the outside world. For example, in image recognition, the input layer receives image pixels.

2. Hidden Layers

Hidden layers perform calculations and extract important features from the input data.

Deep learning models contain multiple hidden layers, which is why they are called "deep" neural networks.

3. Output Layer

The output layer produces the final prediction or result.

Types of Deep Learning Networks

1. Convolutional Neural Networks (CNN)

CNNs are mainly used for processing images and visual data.

Applications:

- Face recognition
- Medical image analysis
- Object detection
- Self-driving cars

2. Recurrent Neural Networks (RNN)

RNNs are designed for sequential data where previous information is important.

Applications:

- Speech recognition
- Time series prediction
- Language modeling

3. Transformers

Transformers are advanced deep learning models that process sequential data efficiently.

Applications:

- Chatbots
- Translation systems
- Large Language Models (LLMs)

Applications of Deep Learning

Deep learning is used in many fields:

Healthcare

Deep learning helps analyze medical images and assist doctors in diagnosis.

Computer Vision

AI systems use deep learning for object detection, image classification, and facial recognition.

Natural Language Processing

Deep learning enables machines to understand and generate human language.

Autonomous Vehicles

Self-driving cars use deep learning for navigation and decision making.

Conclusion

Deep Learning is one of the most powerful areas of Artificial Intelligence. It allows computers to learn complex patterns and solve problems involving images, text, speech, and large datasets.